

MENSCH UND COMPUTER 2026 · STUDENT RESEARCH COMPETITION

Monitoring and Understanding Notebook-Based Learning with Jupyter

A learner-facing JupyterLab dashboard for reflective programming education

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MOTIVATION

The programming process is invisible

Computational notebooks are central to modern data science education. Learners repeatedly test, revise, and debug — but that process leaves no trace in a final submission.

1

Instructors

Lack visibility into how students actually work through a task — only the final notebook is visible.

2

Learners

Rarely get the chance to reflect on their own workflow, error patterns, or problem-solving behavior.

3

Existing dashboards

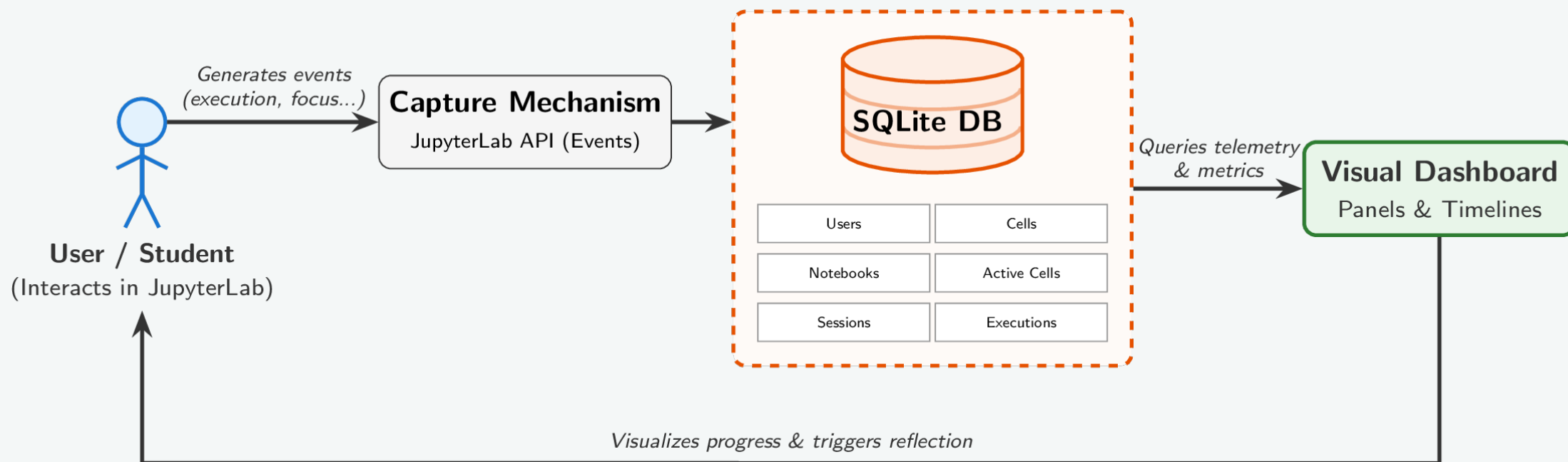
Mostly rely on coarse LMS-level data (grades, access logs) — not fine-grained, cell-level coding behavior.

Research question: Is the dashboard perceived as useful, easy to integrate, and usable — without imposing excessive cognitive workload?

APPROACH

A telemetry pipeline built into JupyterLab

The extension captures notebook interactions live through the JupyterLab API, persists them relationally, and renders reflective visualizations back to the student.

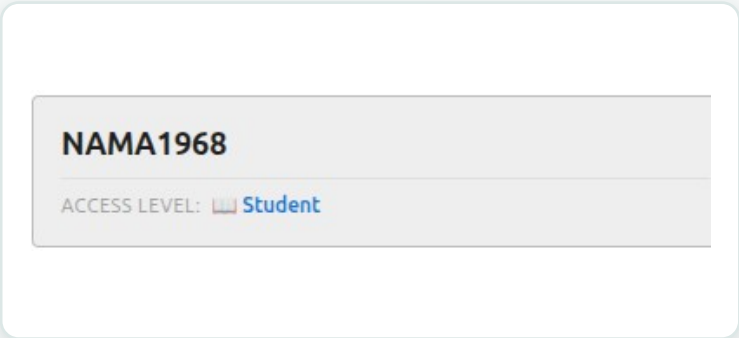


Non-AI, learner-facing reflection interface — embedded directly in the coding environment, not a separate LMS view.

APPROACH

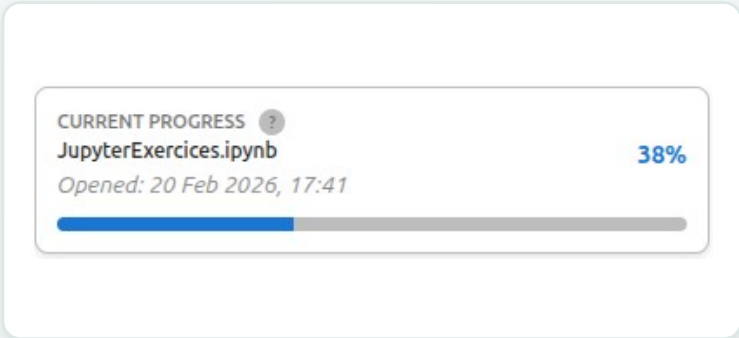
The dashboard, live inside JupyterLab (1/2)

Nine panels, each with its own Help button — accessible from the JupyterLab command palette while students work.



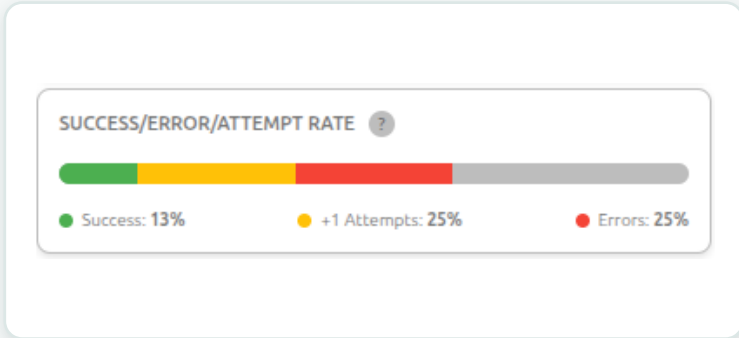
User information

Anonymous ID and role (student / teacher).



Current progress

% of code cells executed without errors.



Success / Error / Attempt rate

Classifies every cell by its final outcome.



Error rate

Failed / total executions, color-graded 0–100%.



Top error cells

Cells with the most failed executions.

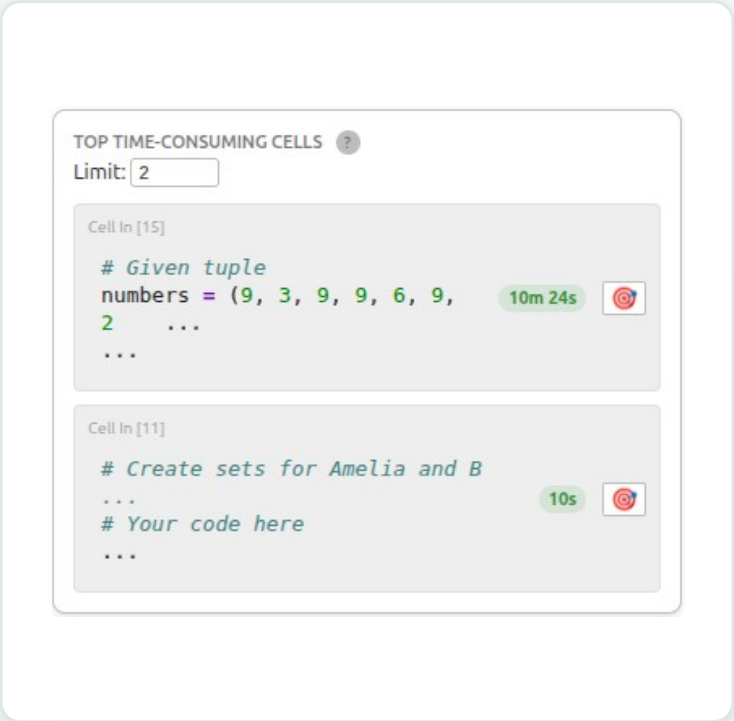


Top attempt cells

Cells with the highest number of executions.

APPROACH

The dashboard, live inside JupyterLab (2/2)



Top time-consuming cells

Cells ranked by total active focus time during the session.



Top common error types

Tally of specific Python exceptions (IndexError, SyntaxError...).



Execution timeline

Chronological, color-coded dot plot of the whole session (green = success, red = error).

Metrics are reflection cues, not diagnostic scores — the Help button explains each one and suggests possible next actions.

An exploratory, learner-centered pilot study

22

students,
varying prior experience

~17 min

average time spent using
the dashboard

3

standardized instruments:
SUS, TAM, NASA-TLX

Procedure

1

Informed consent + background questionnaire (Jupyter / Python experience)

2

Solve Python exercises in JupyterLab while the dashboard runs live

3

Post-use questionnaires: SUS (usability), TAM (usefulness & ease of use), NASA-TLX (workload)

4

Descriptive statistics + exploratory Spearman correlations

RESULTS

Usable and low-workload — usefulness more moderate

65.6

SUS Score

/ 100 · just below the 68 benchmark

5.36

TAM Overall

/ 7 · positive technology acceptance

8.15

NASA-TLX

/ 20 · low-to-moderate workload

Perceived Ease of Use vs. Perceived Usefulness



Students found it easy to operate; its immediate value for self-reflection was less evident — some wanted more actionable, error-specific guidance.

Visibility is only the first step

1

A working system

A JupyterLab extension capturing cell-level execution, error, retry, and focus-time data directly — not coarse LMS logs.

2

Empirical baseline

First learner-centered usability, usefulness, ease-of-use, and workload evidence for a fine-grained notebook dashboard (N = 22).

Implications & next steps

- Visibility alone isn't enough — learners need help interpreting patterns and turning them into concrete actions
- Strengthen the Help function with more actionable, error-specific guidance
- This was a short-term, cross-sectional pilot (~17 min) — it shows usability, not pedagogical impact
- Future work: longitudinal / controlled studies on debugging behavior and learning outcomes over time

Thank you

Questions & discussion

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Video of the Extension



<https://colaps.team/>